

Orch OR Consciousness Theory: Claims, Evidence, Replication, Anesthesia, and Experimental Roadmap

Executive summary

Assessment date: August 22, 2026. Orch OR—“Orchestrated Objective Reduction”—is best understood not as one experimentally established proposition but as a **stack of linked hypotheses**. Roger Penrose supplies the proposed new physics: sufficiently different quantum-superposed mass distributions become unstable because they correspond to incompatible spacetime geometries, with a characteristic reduction time approximately

$$\tau \sim \frac{\hbar}{E_G},$$

where E_G is the gravitational self-energy associated with the difference between the two mass distributions. Stuart Hameroff supplies the proposed biological implementation: neuronal microtubules sustain and “orchestrate” quantum states that reach this objective-reduction threshold; each such reduction is associated with a conscious event, while anesthetics suppress consciousness partly by perturbing the relevant microtubule states. ¹

The most important conclusion of this review is that **several auxiliary premises of Orch OR now have nontrivial experimental support, but the distinctive Orch OR conjunction does not**. Microtubules demonstrably have richer electrical, optical, and collective photophysical properties than a model of them as passive structural beams would suggest. In particular, Kalra et al. measured electronic excitation migration of about 6.6 nm in polymerized microtubules and found that etomidate and isoflurane reduced it; Babcock et al. measured changes in fluorescence quantum yield consistent with theoretical collective superradiant states in hierarchically organized tryptophan networks. These are serious results. Neither experiment, however, demonstrates millisecond-scale coherent quantum computation, entanglement between tubulins or microtubules, an in-brain Orch OR state, or Penrose gravitational collapse. Indeed, Kalra et al. explicitly modeled **incoherent** energy migration as one comparison, and their measurements concern nanometer-scale excitonic transport rather than the long-lived coherent superpositions required by the strongest versions of Orch OR. ²

The anesthesia evidence is more interesting than it was a decade ago but also substantially more ambiguous than strong pro-Orch accounts sometimes imply. Khan et al. reported that acute epothilone B treatment delayed isoflurane-induced loss of righting reflex by 69 seconds in eight rats, with $d = 1.9$. Huang et al. subsequently reported a 29-second within-subject delay in mice, $d = 0.8$, after acute epothilone B. That is a useful cross-species replication **within the same Wellesley research program**, not an independent-laboratory replication. Meanwhile, an independent 2025 mouse study found that two weeks of the related microtubule stabilizer epothilone D shifted the isoflurane LORR curve in the *opposite*

direction—toward greater anesthetic sensitivity—while paclitaxel produced only modest resistance and the destabilizer vinblastine also increased sensitivity. Thus, the general proposition “microtubule stabilization protects consciousness from anesthetic action” is presently falsified in its simplest form; acute versus chronic treatment, specific binding sites, pharmacokinetics, physiology, and drug-specific microtubule effects have to be incorporated into any viable version. ³

The strongest direct molecular bridge between anesthetics and the proposed microtubule mechanism is currently Kalra et al.: 50 μ M etomidate or isoflurane reduced measured microtubule excitation diffusion length from approximately 6.6 nm to approximately 5.6 and 5.8 nm, respectively. But the same paper reports that picrotoxin—a convulsant rather than a general anesthetic—also reduced the relevant quenching behavior in additional experiments. That makes the finding evidence that **microtubule photophysics is pharmacologically modifiable**, but not yet evidence for an anesthetic-specific quantum consciousness switch. ⁴

At the same time, conventional molecular anesthesia mechanisms have much stronger causal evidence than the microtubule hypothesis. A single $\beta 3$ GABA_A-receptor point mutation dramatically attenuates etomidate- and propofol-induced behavioral endpoints in mice; $\beta 2/\beta 3$ manipulations dissociate sedation, hypnosis, amnesia, and immobility; isoflurane acts through both GABA-dependent and GABA-independent thalamic mechanisms; and volatile-anesthetic effects involve multiple ion-channel and circuit targets. These findings do not exclude a microtubule contribution, particularly for volatile anesthetics, but they make a **unitary “anesthetics work primarily by shutting down microtubule quantum coherence” theory much harder to sustain**. ⁵

Claims of genuinely nonclassical brain activity remain intriguing but unsettled. Kerskens and López Pérez reported MRI zero-quantum-coherence signals that they interpreted as an entanglement witness tied to consciousness-related brain activity. The experiment does not identify microtubules as the mediator, has not received an independent exact replication, and its entanglement interpretation was challenged by Warren on NMR grounds; Kerskens and López Pérez published a reply defending their interpretation. It should be listed as a disputed quantum-brain result, **not as established evidence of microtubule entanglement**. ⁶

Penrose's objective-reduction physics is likewise unresolved. The 2021 Gran Sasso experiment by Donadi et al. placed a strong spontaneous-radiation constraint on the natural parameter-free stochastic Diósi–Penrose collapse model, ruling out that implementation. Derakhshani et al. argued that this makes Orch OR based on the simplest gravity-related dynamical collapse model highly implausible. McQueen correctly noted an important qualification: the stochastic model constrained by Gran Sasso is not identical to Penrose's incompletely specified original OR proposal, so the result cannot simply be advertised as “Orch OR disproved.” The appropriate conclusion is narrower: **a major natural implementation of gravity-induced collapse has failed an experimental constraint; Penrose OR itself remains unconfirmed rather than cleanly falsified**. ⁷

My overall evidential judgment is therefore:

Orch OR proposition	Status in 2026	Confidence in assessment
Microtubules have nontrivial electronic/electromagnetic collective dynamics	Supported	High. Multiple experimental approaches show unusual transport, electrical, resonance, or collective optical behavior, although interpretations vary. ⁸
Anesthetics can alter microtubule photophysics	Supported, but presently narrow	Moderate-high for the particular Kalra assay; anesthetic specificity and clinical relevance need replication. ⁹
Manipulating microtubules changes anesthetic susceptibility	Supported as a broad proposition	Moderate. Several animal studies show effects, but direction depends strongly on drug/regimen. ¹⁰
Stabilizing microtubules generically causes anesthetic resistance	Not supported as a general rule	High. Acute EpoB produced resistance; chronic EpoD produced substantially greater sensitivity. ¹¹
Physiological microtubules sustain long-lived coherent/entangled states capable of Orch OR computation	Unproven	High. Existing photophysics does not establish this level of quantum coherence. ¹²
Such states exist in intact brain and correlate specifically with consciousness	Unproven; one disputed non-MT-specific MRI result	High. ¹³
Anesthesia causes unconsciousness specifically by destroying those coherent MT states	Plausible experimental hypothesis, not demonstrated	High. Current evidence connects the endpoints only indirectly. ¹⁴
Penrose gravitational OR occurs according to $\tau = \hbar/E_G$	Unconfirmed; some dynamical realizations strongly constrained	High. ¹⁵
Orch OR as an integrated theory of consciousness	Scientifically articulated but unverified minority hypothesis	High. No experiment yet demonstrates the full causal chain. ¹⁶

One important source-audit point is relevant to the supplied *Orch OR Ascending* primer. It is valuable as an advocacy-oriented map of recent claims, but several statements require tightening before reuse on a claims site. In particular, the statement that Hameroff's quantum-microtubule predictions have now been "confirmed by multiple independent laboratories" is too strong; the rat and 2026 mouse epothilone-B experiments come from the same Wellesley group, and the independent 2025 epothilone-D study goes in the opposite direction. The primer also identifies Saxena et al. 2020 as a *Scientific Reports* paper; it was actually published in *Fractal and Fractional*. And although the 2025 Zoghi et al. conference abstract reports delayed luminescence in tubulin/microtubule constructs, the accessible abstract does not itself substantiate

the stronger claim that anesthetics shortened the delayed-luminescence lifetime. ¹⁷

A concise claims-site formulation that would survive skeptical review is therefore:

Orch OR has acquired experimentally interesting support for some of its subsidiary biological premises—especially nontrivial microtubule photophysics and a possible microtubule contribution to anesthetic action—but there is currently no replicated demonstration of long-lived quantum coherence or entanglement in neuronal microtubules, no established microtubule quantum correlate of conscious state, and no experimental confirmation of Penrose objective reduction.

Theory, claims, and historical development

Orch OR emerged from the convergence of two initially separate programs. Hameroff's early work treated the neuronal cytoskeleton—and especially microtubules—as a possible sub-neuronal information-processing substrate. His 1987 book *Ultimate Computing: Biomolecular Consciousness and NanoTechnology* explicitly included cytoskeletal computing and anesthesia among its major topics. Penrose's 1989 *The Emperor's New Mind* argued that human mathematical understanding could not be adequately captured by ordinary algorithmic computation and suggested that presently unknown physics associated with quantum state reduction might be relevant. His later *Shadows of the Mind* developed the noncomputability argument further while connecting it to quantum-state reduction and the brain. ¹⁸

The decisive synthesis appeared in the joint Hameroff–Penrose paper “Orchestrated reduction of quantum coherence in brain microtubules: A model for consciousness” in 1996. The proposal placed biologically orchestrated quantum states in microtubules at the center of the theory and coupled them to Penrose's gravity-related objective reduction. Penrose's separate 1996 physics paper supplied the characteristic lifetime relation $\tau \sim \hbar/E_G$: superposing two sufficiently different mass distributions means, in his argument, superposing different spacetime geometries, creating an intrinsic ill-definedness in time evolution with energy scale E_G . ¹⁹

The essential Penrose–Hameroff claim structure can be decomposed more rigorously than it often is in popular descriptions.

The computability claim. Penrose argues from Gödel-type considerations that conscious mathematical understanding is not exhausted by execution of a fixed, knowably sound formal algorithm. This is a motivation for seeking nonalgorithmic physics, not an empirical derivation of microtubules. Even a successful Gödel argument would therefore leave open which physical system, if any, implements the relevant noncomputable process. ²⁰

The objective-reduction claim. Ordinary unitary quantum evolution is proposed to be incomplete. A superposition involving sufficiently different mass distributions has gravitational self-energy E_G , with an objective lifetime on the order of $\hbar/E_G$. Unlike Copenhagen-style observer-induced collapse, OR is supposed to be an objective physical event. Penrose's published argument supplies a characteristic criterion rather than a fully specified experimentally complete stochastic dynamical theory, an ambiguity that becomes critical when interpreting experimental bounds on Diósi–Penrose models. ²¹

The microtubule quantum claim. Tubulin proteins in neuronal microtubules are proposed to support quantum states involving conformational, electronic, dipolar, or related degrees of freedom. The 1990s formulation talked more directly about tubulin conformational states as qubit-like units; later formulations increasingly emphasize delocalized dipoles, aromatic electron clouds, collective oscillations, and hierarchical quantum dynamics. ²²

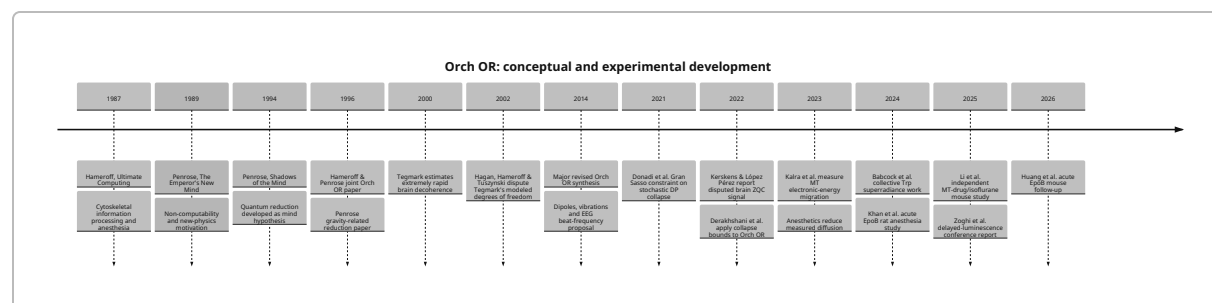
The orchestration claim. These states are not supposed to evolve as an isolated quantum computer. Synaptic inputs, membrane activity, microtubule-associated proteins, cytoskeletal organization, and other biological factors are proposed to “orchestrate” their evolution. The theory consequently claims a bidirectional link: conventional neuronal dynamics organize the microtubule quantum state, while the outcome of reduction subsequently influences neuronal and synaptic behavior. ²³

The conscious-event claim. When the gravitational threshold is reached, objective reduction terminates the quantum computation. Hameroff and Penrose associate each such event with a discrete conscious moment or choice. In the revised 2014 formulation they also proposed that beat frequencies among faster microtubule oscillations could relate the underlying microtubule process to EEG frequencies associated with consciousness. ²³

The phenomenological claim. Orch OR goes farther than most mechanistic neuroscience theories by suggesting that objective reductions access or instantiate primitive experiential qualities associated with fundamental spacetime geometry. This part is metaphysical unless it generates independently discriminating physical predictions; the experimentally actionable components are therefore the microtubule-state, anesthesia, timing, and gravitational-collapse claims. ²³

The anesthesia claim. General anesthetics are proposed to occupy hydrophobic pockets and alter London-dispersion/electronic-dipole behavior within tubulin, thereby interrupting the microtubule quantum process required for consciousness. Later variants connect this idea to collective aromatic-electron/excitonic behavior rather than only to a binary tubulin conformational qubit. ²⁴

The historical trajectory is useful because it shows that “Orch OR” has changed enough that tests must specify **which generation of the theory** is being tested. A finding against one particular tubulin-conformation model does not automatically dispose of a later collective-dipole model; conversely, evidence for ordinary collective exciton transport cannot retrospectively be counted as a demonstration of every earlier Orch OR assertion. ²⁵



The famous decoherence dispute also needs to be represented accurately. Tegmark calculated decoherence times around 10^{-13} – 10^{-20} seconds for the kinds of charged superpositions he modeled, many orders of

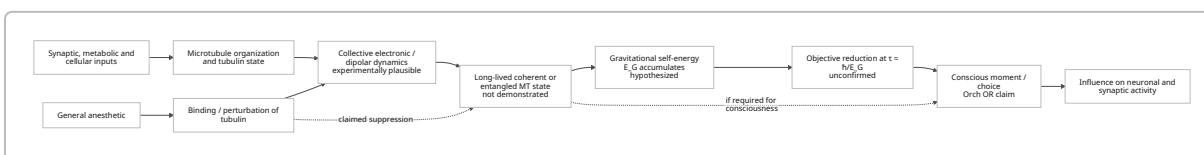
magnitude below neuronal timescales. Hagan, Hameroff, and Tuszynski replied that Tegmark had modeled a different physical superposition than Orch OR intended and, under alternative assumptions, obtained much longer estimates around microseconds to fractions of a millisecond. Neither calculation was a direct measurement of a living-neuron microtubule coherence time. Thus Tegmark did not experimentally refute Orch OR, but neither did the Hagan response experimentally establish biologically useful coherence. ²⁶

This distinction remains central in 2026. The theory needs more than the proposition that “quantum mechanics matters inside proteins,” which is trivially true of all chemistry. The distinctive evidential ladder is:

quantum molecular physics → collective MT excitation → coherent MT superposition → entanglement / nonclass

Experiments presently provide meaningful evidence near the first two stages, scattered and disputed evidence around the fifth, and no accepted direct observation bridging all the way to the final stages. ²⁷

A useful mechanism diagram should therefore visually distinguish **measured** steps from **postulated** ones:



Empirical evidence and replication

The strongest way to assess Orch OR experimentally is to avoid treating all “quantum-looking” observations as equivalent. A collective optical effect, a coherent quantum superposition, entanglement, and gravitational collapse are different physical claims requiring different witnesses. The evidential literature becomes considerably clearer once studies are classified in this way.

Microtubule electronic-energy migration. Kalra et al. 2023 is arguably the most important modern experiment relevant to the biological half of Orch OR. Using intrinsic tryptophan fluorescence lifetimes and concentration-dependent quenching, the group estimated a two-dimensional excitation diffusion coefficient of $3.15 \pm 0.1 \times 10^{-5} \text{ cm}^2/\text{s}$ and a diffusion length of $6.64 \pm 0.1 \text{ nm}$ for polymerized microtubules. Polymerization mattered; 13- versus 14-protofilament architecture made relatively little difference. Conventional Förster calculations incorporating tryptophan and tyrosine were insufficient to reproduce the measured transport. ²⁸

This is genuine evidence for unexpectedly efficient electronic-energy transport across the microtubule lattice. It is **not**, by itself, a demonstration of long-lived quantum coherence. The authors' modeling explicitly includes an incoherent energy-migration model, and the experimentally inferred length scale is approximately one tubulin-dimer dimension. The correct claim is therefore “nontrivial excitonic/electronic transport exists in microtubules,” not “Orch OR quantum computation has been observed.” ⁹

The anesthetic arm of the same experiment is particularly valuable. At 50 μM , etomidate and isoflurane reduced inferred diffusion lengths from approximately 6.6 nm to 5.6 and 5.8 nm. That provides a direct molecular observation linking general anesthetics to a microtubule optical/electronic property. Yet the paper also notes additional experiments in which picrotoxin decreased the same quenching behavior. Picrotoxin is not a general anesthetic, so the present assay has not demonstrated the specificity predicted

by a strong anesthetic-quantum-switch theory. Independent replication across a potency series is now more important than another demonstration with one or two agents. ⁴

Collective tryptophan superradiance. Babcock et al. 2024 modeled very large organized tryptophan networks and predicted highly superradiant states. Experimentally, fluorescence quantum yield increased with hierarchical tubulin organization in the direction predicted by the model. The theoretical architecture involved more than 10^5 transition dipoles and displayed extremely bright and extremely dark collective radiative eigenstates. This is meaningful evidence that geometrically organized biological aromatic networks can display collective quantum-optical behavior at thermal equilibrium. ²⁹

But the distinction between **radiative lifetime** and **quantum information coherence time** is essential. A modeled dark mode with a very long radiative lifetime does not establish that an entangled cognitive state survives environmental dephasing for that duration. Similarly, superradiance establishes cooperative light-matter coupling, not Penrose collapse. Babcock et al. therefore raises the physical plausibility of collective microtubule optics without establishing the neural or consciousness-specific claims of Orch OR. ³⁰

Bandyopadhyay-group electrical and resonance experiments. Sahu et al. reported multilevel electrical memory-switching in single brain-derived microtubules and, in a separate 2013 study, unusual conductivity, fluorescence, vibrational, and water-core-dependent properties. Saxena et al. 2020 later reported “triplet-of-triplet” self-similar electromagnetic resonance structures across isolated tubulin, microtubules, and a neuronal preparation. These experiments are relevant because they challenge the assumption that microtubules are electrically inert scaffolding. ³¹

Their evidential weight for *quantum consciousness*, however, is substantially lower. Resonance, conductivity, memory hysteresis, and scale-free frequency structure are not quantum-coherence witnesses. The highly distinctive resonance claims remain closely associated with one experimental lineage and have not acquired the sort of independent protocol-level replication that would be expected before treating them as established microtubule physics. This is one of the most important unresolved replication gaps in the field. ³²

Delayed luminescence. Zoghi et al.'s 2025 Optica conference contribution reports measurements of delayed luminescence in tubulin and microtubule constructs and states that mesoscale organization alters emission amplitude and decay. This is interesting as another collective-optical probe, but it is conference-proceedings evidence rather than a mature independent replication of the Scholes work. The accessible abstract does not support treating it as a demonstrated anesthetic-dose response, much less as confirmation that anesthetics suppress microtubule “quantum beats” in proportion to clinical potency. ³³

Anesthetic binding and microtubules. Proteomic photolabeling studies have shown that volatile anesthetics interact with many brain proteins rather than one unique receptor. Pan et al.'s halothane-binding proteome work found numerous soluble and membrane targets, consistent with a distributed pharmacology rather than a singular target. That leaves tubulin as a candidate target but weakens arguments that Meyer–Overton behavior logically demands one unique microtubule site. ³⁴

Emerson et al. 2013 provided stronger causal evidence for a tubulin-anesthetic interaction in a particular chemical system. A photoactive anthracene anesthetic labeled β -tubulin near the colchicine-binding region; crosslinking affected polymerization, and microtubule-stabilizing agents shifted the concentration needed to immobilize tadpoles. This experiment establishes that tubulin can lie on an anesthetic-relevant pathway

for at least some compounds and behavioral endpoints. It does not demonstrate that clinical volatile-anesthetic unconsciousness is mediated through an Orch OR quantum state. ³⁵

Craddock et al. 2017 is sometimes cited as “experimental” evidence for the anesthesia arm of Orch OR, but it should be classified primarily as computational/theoretical work. Molecular docking and mode calculations suggested that anesthetic interactions with tubulin could alter collective dipole modes in a way correlated with anesthetic potency. The study explicitly called for experimental confirmation. It generates testable predictions; it does not itself supply that confirmation. ³⁶

Acute epothilone-B behavioral experiments. Khan et al. 2024 used eight male Long-Evans rats in the main analysis, measuring LORR under 4% isoflurane before and after a 0.75 mg/kg subcutaneous dose of brain-penetrant epothilone B. Seven of eight rats showed increased average latency, the group mean increased by 69 seconds, $p = 0.0016$, and Cohen's $d = 1.9$. Experimenters making LORR measurements were blinded; an additional four-rat group was used to investigate tolerance from repeated anesthesia. ³⁷

The result deserves attention, but its inference is narrower than the article's strongest wording. The main sample is tiny; every animal receives the critical EpoB phase after the pre-EpoB phase rather than through a fully counterbalanced placebo/EpoB crossover; LORR is an indirect behavioral proxy; and EpoB alters microtubule stability and many downstream cellular processes. Most importantly, the experiment measured no microtubule quantum variable. It establishes a causal relationship between acute EpoB treatment and altered isoflurane susceptibility under that protocol, not between “quantum microtubule coherence” and consciousness. ³⁸

Huang et al. 2026 reproduced the direction of this effect in mice after a single 8 mg/kg EpoB injection: LORR latency the next day increased by 29 seconds within subjects, with $d = 0.8$, and the effect weakened on subsequent days. This is valuable cross-species evidence. It is not independent replication because the senior author and research program are the same as in the rat experiment. ³⁹

The independent contradictory mouse result is essential. Li et al. 2025 randomly assigned male CD1 mice to prolonged microtubule-modulating treatments and measured stepwise isoflurane LORR curves under blinded testing. After two weeks of epothilone D, the LORR EC50 moved from approximately 0.97% in saline controls to 0.75%, indicating **increased**, rather than decreased, isoflurane sensitivity. Vinblastine likewise produced greater sensitivity; paclitaxel produced a small shift toward resistance. The EpoD experiment used 12 animals per group. ⁴⁰

The authors themselves identify important limitations: saline was not an ideal vehicle control for the poorly soluble agents; LORR rather than MAC was used; only isoflurane was tested; outbred CD1 animals contributed variability; and EpoD altered blood pH and CO₂. Those caveats make the result less decisive, but they do not make it disappear. A credible Orch OR synthesis has to include it because it demonstrates that the relation between “microtubule stabilization” and anesthetic resistance is not monotonic or drug-independent. ⁴¹

Xenon isotope experiment. Li et al. 2018 randomized 80 male C57BL/6 mice across xenon isotopes differing in nuclear spin and reported that spin-bearing ¹²⁹Xe and ¹³¹Xe were less potent in the LORR assay than spin-zero isotopes, despite essentially identical ordinary electronic chemistry. This is legitimately intriguing because nuclear spin is a quantum degree of freedom. But the experiment does not locate the relevant interaction in microtubules, and xenon was tested with 0.50% isoflurane because of xenon's low

potency. Spin-dependent biochemical mechanisms elsewhere in the nervous system remain possible. It therefore raises the prior probability of a quantum-sensitive anesthetic mechanism more than it specifically confirms Orch OR. ⁴²

MRI “entanglement” evidence. Kerskens and López Pérez used a zero-quantum-coherence NMR protocol intended to minimize conventional MRI contrasts and reported widespread brain signals with heartbeat-evoked timing whose presence depended on conscious awareness. They argued that the protocol could act as an entanglement witness and concluded that consciousness-related brain function might be mediating entanglement. ⁴³

This result sits much higher on the Orch OR evidential ladder than ordinary exciton transfer *if* the interpretation is correct. But it is correspondingly more controversial. Warren argued that the data do not establish entanglement and that complex classical NMR behavior in heterogeneous tissue undermines the proposed witness interpretation. Kerskens and López Pérez replied that the reported signal cannot be explained by the specific classical intermolecular multiple-quantum-coherence mechanism invoked in the criticism. No independent exact replication establishing an accepted entanglement witness in living human brain has emerged in the literature reviewed here. Nor does the experiment implicate microtubules specifically. ⁴⁴

Objective-reduction experiments. Donadi et al. conducted a conceptually different and exceptionally important test at Gran Sasso. A stochastic Diósi–Penrose collapse model predicts momentum diffusion and therefore spontaneous electromagnetic radiation from charged particles. The experiment searched for the predicted radiation and did not find it, tightening the effective nuclear-size bound by about three orders of magnitude and ruling out the model's natural parameter-free form. ⁴⁵

Derakhshani et al. subsequently examined the implications for Orch OR and concluded that versions based on the simplest gravity-related dynamical collapse implementation were highly implausible. McQueen's response makes the crucial model-theoretic point: a stochastic DP master equation is a particular completion of gravity-induced collapse, whereas Penrose's original OR proposal is not fully equivalent to it. The Gran Sasso result is therefore a **strong null result against an important implementation**, not a universal empirical disproof of every physics theory that could be called Penrose OR. ⁴⁶

The overall literature has not reached a state in which a conventional statistical meta-analysis is very meaningful. The main studies measure different quantities—nanometer excitation diffusion, fluorescence quantum yield, electrical impedance, LORR latency, LORR EC50, NMR signals, and spontaneous radiation—and several apparently separate positive studies are not independent because they share investigators or laboratory programs. I found no mature systematic meta-analysis pooling Orch OR experimental effects; the appropriate synthesis today is therefore **claim-by-claim replication analysis**, not a single pooled effect size. ⁴⁷

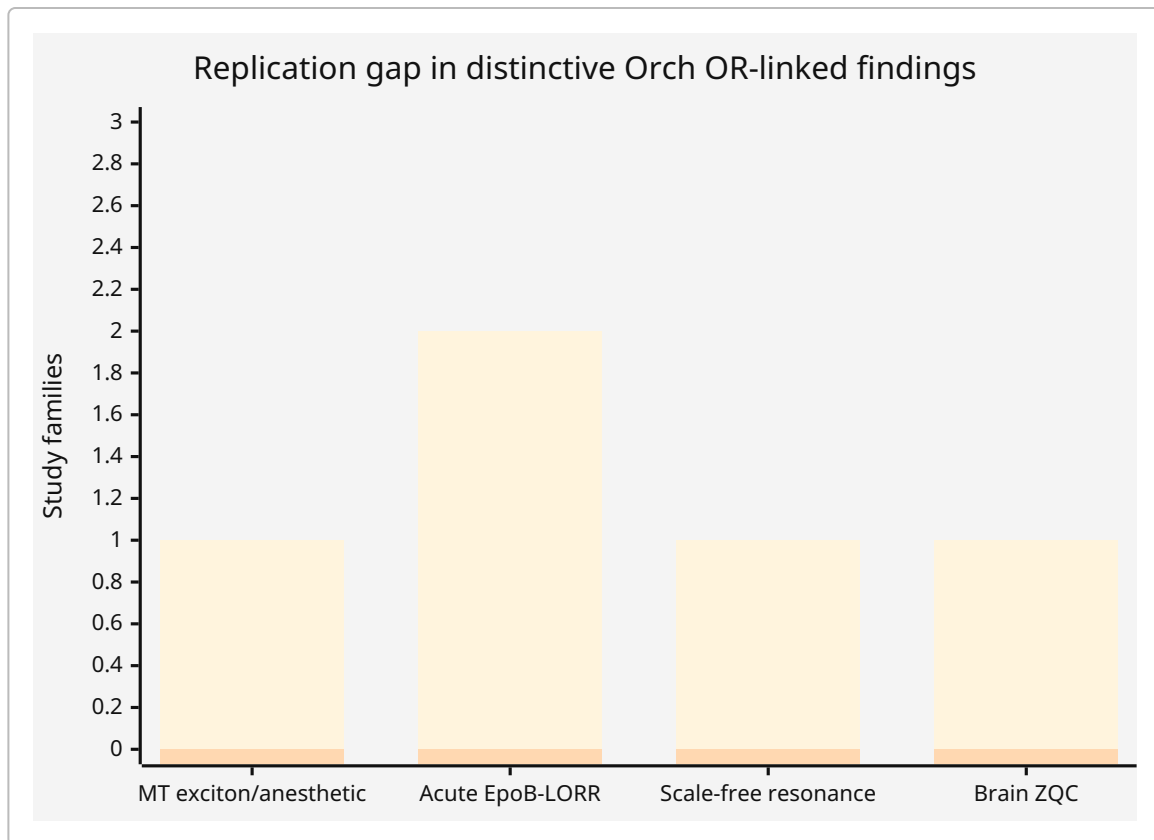
The following table grades **reliability of the measured result and diagnosticity for Orch OR**, not prestige of the journal. “A” means unusually strong for the specific question; “B” methodologically useful with important limitations; “C” suggestive or interpretation-dependent; “D” currently insufficient as a discriminator.

Study	Method / sample	Main result	Exact or meaningful replication status	Orch OR relevance	Rating
Tegmark, 2000	Open-system decoherence calculation	Very short decoherence times under modeled charge-superposition assumptions	Hagan et al. disputed model assumptions; no direct measurement	Important theoretical challenge, not experiment	B for challenge, indirect
Hagan, Hameroff & Tuszynski, 2002	Revised decoherence calculations	Longer potential MT coherence times under different physical assumptions	No direct experimental confirmation of those lifetimes	Shows Tegmark result is model-dependent	B-/ 48
Sahu et al., 2013	Single-MT electrical/optical/AFM/fluorescence measurements	Unusual conductivity, water-core dependence and size-independent features	No broadly established independent protocol replication	Supports active MT electrodynamics, not coherence	C+ 49
Emerson et al., 2013	Photoreactive anesthetic, tubulin labeling/polymerization, tadpole behavior	β -tubulin implicated in anthracene anesthesia	Related but not exact later work	Direct evidence that tubulin can be anesthetic-relevant	B 35
Saxena et al., 2020	Impedance/transmission/resonance across tubulin, MT, neuron	Self-similar resonance patterns	No independent exact replication identified	Interesting but not a quantum witness	C 50
Donadi et al., 2021	Underground germanium radiation experiment	No predicted DP radiation; parameter-free stochastic version excluded	Strong experimental constraint	Directly relevant to one OR implementation, not MT biology	A 45

Study	Method / sample	Main result	Exact or meaningful replication status	Orch OR relevance	Rating
Kerskens & López Pérez, 2022	Human-brain ZQC MRI	Signals interpreted as consciousness-related entanglement	Interpretation challenged; no independent exact replication	Potentially major if validated; currently disputed	C ¹³
Kalra et al., 2023	Trp fluorescence lifetime/ quenching in purified MTs	~6.6 nm excitation migration; isoflurane and etomidate reduce transport	No independent exact replication yet	Strongest direct MT photophysics/ anesthetic bridge	B+ ⁹
Babcock et al., 2024	Quantum-optical modeling + fluorescence-QY measurements	Collective superradiant behavior consistent with hierarchical Trp networks	Independent adjacent evidence, not exact Kalra replication	Supports collective MT optics, not long-lived cognitive coherence	B ⁵¹
Khan et al., 2024	N=8 main rat within-subject experiment, EpoB + isoflurane LORR	+69 s LORR latency, $d = 1.9$	Same-group mouse follow-up; no independent exact replication	Causal MT-drug/ anesthesia evidence	B- ⁵²
Li et al., 2025	Randomized/ blinded male CD1 mice, multiple MT drugs	EpoD and vinblastine increased sensitivity; paclitaxel slight resistance	Independent of Wellesley; direction partly contradictory	Essential constraint on simple stabilization hypothesis	B ⁵³
Zoghi et al., 2025	Delayed-luminescence measurements	Mesostructure alters emission decay	Conference-level; no exact independent replication established	Interesting optical lead, currently preliminary	C- ³³

Study	Method / sample	Main result	Exact or meaningful replication status	Orch OR relevance	Rating
Huang et al., 2026	Acute EpoB, mouse within-subject LORR	+29 s next-day latency, $d = 0.8$	Replicates direction across species but same research group	Strengthens EpoB phenomenon, not quantum mechanism	B- 39

A useful replication visual should distinguish “another positive paper” from an **independent exact replication**. For four of the most distinctive Orch-OR-linked observations, the latter remains the major missing commodity. The first bar series below represents published positive study families or same-program follow-ups; the second represents independent exact replications identified in this review.



That chart should **not** be read as “all evidence is unreplicated”: Babcock supplies independent *adjacent* support for collective microtubule optical phenomena, and Li et al. independently confirm that microtubule-modulating drugs can alter anesthesia. What is lacking is replication of the specific signatures that are being used as distinctive confirmation of Orch OR. 54

Anesthesia claim: practical evaluation

The anesthesia hypothesis is currently the experimentally strongest and most tractable component of Orch OR because it can be separated from both Penrose's Gödel argument and the harder objective-collapse physics. Even if consciousness has nothing to do with gravitational collapse, a discovery that neural microtubules are major anesthetic targets would be important. Conversely, demonstrating that anesthetic effects on microtubules are epiphenomenal would remove one of Orch OR's central biological supports.

The hypothesis needs to be stated at several strengths.

Weak claim: clinically relevant anesthetics interact with tubulin/microtubules and alter their function. This is increasingly credible. Protein-binding studies, anthracene/tubulin work, Kalra's photophysics, and microtubule-drug behavioral experiments collectively support it. ⁵⁵

Intermediate claim: alteration of microtubules makes a causal contribution to anesthetic hypnosis or unconsciousness. This is plausible but not yet securely established. EpoB/EpoD/paclitaxel/vinblastine experiments show that changing microtubule biology changes isoflurane susceptibility, but these drugs have broad consequences, and the direction is inconsistent across stabilizers and regimens. A microtubule contribution need not be quantum mechanical. ¹⁰

Strong Orch OR claim: anesthetics cause unconsciousness substantially by disrupting a specific quantum-coherent microtubule state that is itself required for consciousness. No existing experiment directly establishes this. No study has simultaneously measured a validated MT quantum-coherence variable, shown anesthetic-dose-dependent suppression of that variable, rescued it independently of ordinary cellular physiology, and shown that rescue preserves consciousness. ⁵⁶

The proposed molecular mechanism usually invokes hydrophobic pockets in tubulin and the influence of anesthetics on polarizable aromatic electron clouds. London-dispersion interactions could change tubulin's collective dipole/excitonic spectrum, shortening or damping whatever coherent state is supposed to participate in Orch OR. Kalra provides the first compelling direct bridge to part of this model because isoflurane and etomidate alter measured excitation migration. Craddock's earlier calculations supply a more specific hypothesis that anesthetic potency should correlate with changes in collective tubulin modes, but the latter remains a modeling prediction requiring prospective experimental validation. ²⁴

The Meyer–Overton correlation does not uniquely select this mechanism. Lipid solubility historically suggested a hydrophobic site, but hydrophobic cavities occur in many proteins, and modern genetic experiments prove that specific protein receptors can mediate large fractions of particular anesthetic endpoints. The $\beta 3(N265M)$ GABA_A mutation essentially abolishes etomidate- and propofol-induced immobility while profoundly reducing hypnotic effects, providing much cleaner target-level causality than presently exists for tubulin. ⁵⁷

The story is more distributed for volatile agents. Isoflurane modulates thalamic activity through GABA_A-dependent as well as GABA-independent mechanisms, and altered voltage-gated sodium-channel function changes sensitivity to isoflurane and sevoflurane. TREK-1 was initially attractive because knockout mice were reported to resist volatile anesthesia, but later work examining TREK-1/TREK-2 knockouts did not

reproduce the simple resistance result. This history is instructive: even apparently strong anesthetic-target claims can change after genetic replication. ⁵⁸

Accordingly, the most defensible present model is **multi-target anesthesia**, with the possibility that microtubules are one contributing target whose importance may depend on agent, concentration, cell type, and endpoint. Etomidate and propofol have unusually strong GABA_A-receptor evidence; volatile anesthetics act more broadly. A successful microtubule theory does not need to replace these mechanisms, but it has to demonstrate an effect that cannot simply be reduced to altered ion-channel signaling, metabolism, axonal transport, cytoskeletal integrity, or pharmacokinetics. ⁵⁹

The current epothilone evidence is particularly useful for generating discriminating predictions. Acute EpoB causes resistance in the Wellesley rat and mouse experiments. Chronic EpoD instead produces strong hypersensitivity in the independent Massachusetts General Hospital study. Possible explanations include different binding sites, EpoB versus EpoD pharmacology, acute versus chronic microtubule remodeling, compensatory neuronal changes, blood-gas effects, sex, genetic background, and nonlinear dependence on microtubule stability. Those possibilities are scientifically reasonable—but every added degree of freedom reduces the predictive content of the original simple claim. A mature Orch OR anesthesia model must specify **before the experiment** which manipulations should increase versus decrease anesthetic potency. ⁶⁰

The single best near-term experiment would therefore not be another uncontrolled microtubule stabilizer study. It would be a **causal tubulin-binding-site experiment**.

The program would first use photolabeling, quantitative mass spectrometry, cryo-EM, and atomistic calculations to map a reproducible isoflurane/etomidate-sensitive tubulin pocket. Investigators would then engineer one or more conservative tubulin substitutions predicted to greatly reduce anesthetic occupancy while preserving polymerization kinetics, microtubule dynamic instability, motor transport, and neuronal viability. Conditional neuronal knock-in mice would avoid developmental confounding. Anesthetic brain concentration would be measured directly so that pharmacokinetics could not explain altered behavior. Endpoints would include equilibrium LORR EC50, MAC for immobility, EEG complexity, sensory-evoked cortical responses, induction and emergence kinetics, body temperature, arterial gases, and motor performance. ⁶¹

A strong Orch-compatible prediction would be that disrupting the tubulin anesthetic pocket produces selective anesthetic resistance **and simultaneously prevents the anesthetic-induced microtubule photophysical change**, while leaving baseline neuronal physiology and established GABA_A responses essentially intact. Re-expression of wild-type tubulin should restore both photophysical suppression and behavioral susceptibility. A mutation that changes the photophysical signal without changing anesthesia would argue that the optical phenomenon is epiphenomenal. A mutation that changes anesthesia without changing the proposed quantum signal would argue for an ordinary microtubule mechanism rather than Orch OR. This triangulation is far more diagnostic than another correlation.

A second decisive experiment would test the potency prediction prospectively. Purified neuronal microtubules should be exposed, under identical conditions, to a panel of perhaps 8–12 anesthetics covering volatile agents, intravenous agents, xenon, structurally related “nonimmobilizer” compounds, and nonanesthetic hydrophobic controls. Free aqueous concentration—not nominal stock concentration—must be measured. The prespecified prediction should be that suppression of a **validated coherent**

spectroscopic observable tracks clinical potency across compounds and concentrations. A correlation chosen only after inspecting the data would have much less evidential force.

A third decisive test would ask whether the putative microtubule state precedes and predicts the behavioral transition. In neuronal cultures, brain slices, organoids, and ultimately animals, investigators would simultaneously measure electrophysiology and the best available MT-specific optical/coherence signal during graded anesthetic wash-in and wash-out. If the MT variable is causal, its loss should precede network collapse in a reproducible concentration-dependent manner, and selective restoration of the MT variable should delay or prevent the downstream state transition. If it follows loss of firing, it is more likely a downstream consequence.

The xenon isotope result offers a useful auxiliary test. The isotope effect should be replicated independently, preferably without background isoflurane if a technically feasible endpoint can be developed. It should then be tested in tubulin-binding-site mutants. If the nuclear-spin effect disappears when the predicted tubulin interaction is disrupted, the result would become dramatically more specific to Orch-compatible quantum anesthesia. If it persists, another spin-sensitive molecular mechanism is implicated. ⁴²

A practical evidence score for the anesthesia chain is:

Link in the causal chain	Current support	What is missing
Anesthetic molecules interact with tubulin/MTs	Moderate-strong	Quantitative site occupancy at clinical brain concentrations across anesthetic classes. ⁶¹
Anesthetics alter MT electronic/optical behavior	Moderate	Independent replication, dose-response, potency series, nonanesthetic controls. ⁹
MT manipulation changes anesthetic behavior	Moderate but heterogeneous	Multicenter replication and mechanistic explanation of opposite stabilizer results. ¹⁰
A specific MT optical change causes LORR/unconsciousness	Weak	Simultaneous signal + behavior + rescue experiment
The relevant MT signal is genuinely quantum coherent/entangled	Very weak / unproven	Direct coherence/entanglement witness
That quantum state is necessary for consciousness outside anesthesia	Unproven	Wake/sleep/perception experiments with MT-specific quantum readout
Anesthetic suppression prevents Penrose OR	Unproven	Requires independent confirmation of both MT state and OR physics

Scientific consensus and open controversies

There is no authoritative poll or academy statement establishing a numerical “consensus percentage” on Orch OR. The most defensible characterization comes from the pattern of evidence: **Orch OR remains outside mainstream mechanistic neuroscience, anesthesia science, and quantum physics as an accepted theory, while surviving as a minority research hypothesis with an increasingly empirical experimental program.** The recent experiments make it less reasonable to dismiss the entire subject merely with the phrase “the brain is warm and wet”; they do not make it reasonable to say that quantum microtubule consciousness has been demonstrated. ⁶²

The 2014 *Physics of Life Reviews* exchange remains representative of this divide. Hameroff and Penrose presented a revised Orch OR model incorporating developments in quantum biology, neuroscience, vibrations, and anesthetic mechanisms. Reimers and colleagues responded that the revised proposal was still not scientifically justified, while other commentators focused on whether biological environments necessarily preclude quantum dynamics. The subsequent decade has strengthened some of the biological premises but has not resolved the core dispute. ⁶³

The largest conceptual controversy is **what kind of quantum effect is actually required.** Almost all molecular biology is quantum mechanical at some microscopic level. Orch OR becomes distinctive only when it requires a state that classical or semiclassical molecular dynamics cannot adequately reproduce and that survives long enough, extends far enough, and couples strongly enough to affect cognition. Efficient excitation hopping and collective fluorescence are therefore relevant but insufficient. Without a quantitative threshold specifying coherence length, coherence time, participating degrees of freedom, and required entanglement structure, virtually any new microtubule photophysics can be absorbed into the theory after the fact. ¹²

A second controversy is **decoherence.** Tegmark's original calculation was not the last word because its chosen superposition variables and electrostatic assumptions do not uniquely describe every plausible microtubule quantum state. But substituting a different calculation is not equivalent to observing the proposed state. The debate can now be settled experimentally: specify the state, prepare or identify it, measure its off-diagonal coherence or a suitable witness, and determine T_2 /dephasing under physiological conditions. Until this occurs, both sweeping “instant decoherence kills Orch OR” and sweeping “warm quantum biology solves decoherence” claims go beyond the data. ²⁶

A third controversy is **scale.** The Kalra excitation diffusion length is of order nanometers, while consciousness requires integration across vastly larger neural architectures. Orch OR therefore requires a bridging mechanism from local molecular collective effects to extended states across many tubulins, potentially multiple microtubules and neurons. The Saxena/Bandyopadhyay scale-free resonance program is intended to provide such a hierarchy, but resonance-frequency similarity is not evidence of quantum entanglement across those scales. ⁶⁴

A fourth controversy is **how microtubules couple to ordinary electrophysiology.** Even incontrovertible quantum processing inside a microtubule would be biologically irrelevant unless its output can alter ion channels, synapses, cytoskeletal transport, membrane excitability, or network dynamics on useful timescales. Orch OR proposes such feedback, but the distinctive quantum-to-classical transduction pathway has not been isolated. This is a stronger criticism than merely observing that microtubules have

conventional cellular functions: any quantum theory has to demonstrate an incremental causal effect beyond those conventional functions. ²³

A fifth controversy is **what the anesthesia experiments actually measure**. LORR is useful, reproducible, and conventional in animal anesthesia research, but it is not identical to phenomenal unconsciousness. Anesthetic endpoints fractionate: amnesia, hypnosis, immobility, analgesia, autonomic suppression, and cortical integration can involve different receptors and circuits. Genetic GABA_A experiments explicitly dissociate these components. Consequently, a drug-induced change in LORR does not by itself identify the physical substrate of conscious experience. ⁶⁵

A sixth controversy concerns **Penrose OR versus Diósi–Penrose collapse models**. The formula $\tau \sim \hbar/E_G$ motivates both, but the mapping from Penrose's heuristic geometric instability to a particular stochastic master equation is not unique. Donadi's null result is devastating to a natural stochastic DP realization because that theory makes an extra radiation prediction. McQueen is therefore right that one cannot logically substitute “one DP model is excluded” for “every possible Penrose OR theory is excluded.” The converse is equally important: escaping the experimental bound by retreating to an incompletely specified version of OR does not count as positive evidence for that version. ⁶⁶

Independent gravity-related collapse experiments are nevertheless one of Orch OR's virtues compared with theories whose fundamental mechanism cannot be tested outside consciousness research. Howl, Penrose, and Fuentes have proposed Bose–Einstein-condensate experiments in which a sufficiently massive spatial superposition should decay according to the Penrose gravitational self-energy criterion. A clean positive OR signal would validate a central physical ingredient of Orch OR while still leaving the biological microtubule implementation to be demonstrated separately. ⁶⁷

The current field can therefore be represented as three partly independent questions:

Question	Scientific status
Can biological structures support nontrivial room-temperature quantum/collective dynamics?	Yes. This is no longer controversial in the broad sense, and microtubules themselves show interesting collective optical behavior. The relevant question is <i>which</i> quantum dynamics and for how long. ¹²
Do neuronal microtubules implement consciousness-relevant coherent quantum information processing?	Unknown. No accepted direct demonstration exists.
Does gravity objectively reduce wavefunctions according to the Penrose criterion?	Unknown. Important stochastic implementations are experimentally constrained; the underlying Penrose idea remains unconfirmed. ⁶⁸

Recent pro-Orch syntheses should be read in that context. Wiest's 2025 *Neuroscience of Consciousness* article makes a forceful affirmative case from anesthesia, microtubule quantum optics, and the unity/binding problem. Sergi et al. 2025 attempt to sharpen the quantum–classical architecture and explicitly acknowledge that inter-neuronal microtubule entanglement has not been demonstrated. These papers

show that Orch OR is an active research program, not that its interpretation has become community consensus. 69

For a claims site, the most important terminology discipline is therefore:

“Quantum effect in microtubules” ≠ “quantum coherence in microtubules” ≠ “entanglement in microtubules” ≠ “quantum computation in neurons” ≠ “Orch OR” ≠ “Penrose gravitational collapse.”

Each implication requires a separate experiment. Much of the public controversy arises from silently moving across those equals signs.

Experimental roadmap

The next generation of Orch OR research should be designed around **discriminating predictions rather than demonstrations of generic microtubule complexity**. The field already knows that microtubules are complex. The question is whether a specifically nonclassical microtubule state exists, whether anesthesia selectively suppresses it, whether it causally participates in consciousness, and whether Penrose OR supplies its terminating dynamics. The following program is ordered by information gain per unit cost.

Priority	Experiment	Core prediction	Essential controls	Technology / feasibility	Realistic timeline
Highest	Independent multicenter acute-EpoB anesthesia replication	Acute brain-penetrant EpoB reproducibly shifts isoflurane dose-response toward resistance	Randomized counterbalanced vehicle/EpoB; both sexes; blood gases; temperature; brain anesthetic concentration; baseline motor testing; multiple sites	Standard animal anesthesia + pharmacology; high feasibility	12–18 months
Highest	Prospective anesthetic-potency vs MT photophysics series	A prespecified MT quantum/optical metric is suppressed in proportion to anesthetic potency	≥ 8 agents; nonimmobilizers; picrotoxin; hydrophobic nonanesthetics; blinded analysis; measured free drug concentration	TCSPC, ultrafast spectroscopy, 2DES; high-moderate feasibility	1–2 years

Priority	Experiment	Core prediction	Essential controls	Technology / feasibility	Realistic timeline
Very high	Tubulin anesthetic-pocket knock-in	Disrupting MT anesthetic binding causes behavioral resistance and preserves the MT signal	Polymerization/transport controls; rescue allele; established GABA target controls; PK measurement	Cryo-EM/MS + CRISPR conditional mouse; moderate feasibility	3–5 years
High	Direct MT coherence spectroscopy at 37 °C	Reproducible nonclassical coherence persists beyond classical population dynamics and is anesthetic-sensitive	Isotope substitution, temperature series, denatured protein, disorder controls, independent analysis	2D electronic spectroscopy / pump-probe / photon-echo; moderate	1–3 years
High	Live-neuron MT state + electrophysiology	MT signal changes before anesthetic loss of network integration and predicts recovery	Nonanesthetic activity suppression; GABA blockade; MT-specific perturbations	Ultrafast microscopy, MEA/patch clamp, neuronal cultures/organoids	2–4 years
Medium-high / high reward	MT-specific entanglement witness in cells	A Bell-, spin-squeezing-, or other validated nonclassical witness is observed and abolished specifically by MT disruption	Classical correlated-noise models; blind synthetic controls; independent lab	Quantum sensing / advanced NMR / NV magnetometry; difficult	4–7 years
Independent physics track	Direct gravity-induced collapse test	Superposition lifetime follows $\hbar/E_G$, distinguishable from environmental decoherence	Mass/separation scaling; vibration/thermal controls; alternative collapse models	BEC or levitated/nanomechanical systems; very difficult	~5–10+ years

Independent multicenter EpoB replication. The first priority is intentionally mundane. A phenomenon should be made boring before it becomes foundational. Three or more anesthesia laboratories with no prior stake in Orch OR should preregister a common protocol. Animals should be randomized to acute EpoB or matched vehicle and, where biologically feasible, crossed over in counterbalanced order. Investigators should quantify complete isoflurane LORR concentration-response curves rather than only latency at 4%, and include MAC or another immobility endpoint to determine whether the effect is specific to hypnosis-like behavior. Both sexes are essential given the 2026 paper's suggestion of sex-related differences. ⁷⁰

A decisive replication criterion could be specified in advance: the pooled multicenter EpoB effect must move the LORR EC50 toward resistance with the same sign at every site and survive adjustment for temperature, arterial gases, EpoB plasma/brain level, isoflurane concentration, baseline motor performance, and repeated-exposure tolerance. EpoD and paclitaxel should be included to determine whether the contradictory 2025 pattern is a regimen effect or a compound-specific effect. ⁵³

Prospective potency-series spectroscopy. The field presently has what is essentially an $n = 2$ anesthetic demonstration in the strongest direct microtubule-optical study: etomidate and isoflurane. The next experiment should preregister a quantitative relationship such as

$$\Delta C_{MT}(c) \propto f(\text{anesthetic potency}),$$

where C_{MT} is a physically defined coherence amplitude, diffusion coefficient, dephasing time, or spectral cross-peak rather than an ad hoc signal extracted after inspection. Isoflurane, sevoflurane, desflurane, halothane where practical, xenon, propofol, etomidate, and perhaps ketamine should span distinct chemistries. Structurally related nonimmobilizers and picrotoxin are indispensable because Kalra's picrotoxin observation shows that simple suppression is not anesthetic-specific. ⁹

The experiment becomes much stronger with two-dimensional electronic spectroscopy. Population transfer and coherent oscillatory cross-peaks can then be separated through waiting-time dependence and global kinetic modeling. Isotopic substitution, controlled deuteration, temperature variation, and mutation of aromatic residues can distinguish electronic from vibrational/vibronic coherence. A genuine Orch-relevant result would require more than a long fluorescence lifetime: it would require a defined phase-coherent observable with a lifetime and spatial scale sufficient for the proposed computation.

Tubulin binding-site genetics. This is probably the single experiment with the greatest long-run discriminating power. In pharmacology, a mutation that selectively removes a drug's effect at a candidate protein while leaving the protein functional is vastly more informative than systemic administration of a broad stabilizer. GABA_A anesthesia genetics demonstrates why: the N265M $\beta 3$ mutation converted receptor pharmacology into causal in-vivo evidence. Orch OR needs the equivalent experiment for tubulin. ⁵⁷

The technical challenge is tubulin redundancy and essentiality. A realistic design would target a dominant neuronal β -tubulin isotype or introduce the mutation conditionally in adult forebrain/thalamocortical neurons. Before animal work, recombinant mutant tubulin must retain normal polymerization, GTPase activity, lattice parameters, kinesin/dynein transport, and baseline photophysics while showing strongly reduced anesthetic occupancy. Failure to preserve these functions would make any behavioral result uninterpretable.

A **positive** result would be unusually compelling if three phenomena co-segregated: reduced anesthetic binding to mutant tubulin, reduced anesthetic suppression of the candidate MT quantum/optical signal, and higher anesthetic concentration required for equivalent conscious-state disruption. A rescue transgene restoring wild-type tubulin should restore all three.

Direct coherence measurements. The most damaging ambiguity in Orch OR research is the habitual substitution of “resonance,” “energy migration,” or “superradiance” for “coherence.” The field should specify a density-matrix observable. Depending on the proposed degree of freedom, useful measurements include photon-echo decay, 2DES cross-peak oscillations, Ramsey-type spectroscopy, spin coherence T_2 , or a validated entanglement witness. The target is not an indefinitely long state; it is the timescale explicitly required by a quantitative Orch OR model.

A useful experiment would compare purified tubulin, polymerized microtubules, microtubules plus physiological MAPs, cytosol-like crowded solution, isolated axons, and live neurons. If polymerization creates a protected coherence, its T_2 should increase reproducibly in the ordered lattice and show a predictable response to structural disorder. If it collapses as soon as physiological ionic strength, MAP binding, molecular crowding, or 37 °C temperature is introduced, the strongest neural version of the theory would be severely constrained.

Neuron-level causal coupling. Even a positive coherence result must cross the quantum-to-neural gap. The key measurement is temporal precedence. During anesthetic wash-in, simultaneous MT spectroscopy and membrane/network electrophysiology should determine whether the MT state disappears before cortical-like network integration, simultaneously with it, or after it. A selectively MT-binding perturbation that removes the signal without grossly disturbing microtubule structure should test necessity.

The control condition should suppress neural activity conventionally—through ion-channel manipulation, cooling, or metabolic inhibition—without directly binding the proposed tubulin pocket. If the MT “quantum” signal simply disappears whenever neurons stop firing, it may be a consequence rather than a driver of neural activity. Conversely, persistence of the signal during ordinary electrical suppression but selective disappearance during loss of conscious-like network integration would be much harder to explain classically.

Validated entanglement witnessing. A next-generation version of the Kerskens program would be important if it explicitly separates quantum entanglement from classical multiple-quantum NMR signals. The validation should first be performed in engineered phantoms where the ground truth is known, then in ex vivo tissue, then in living brain. Independent groups should reproduce the pulse sequence and raw signal processing before consciousness correlations are interpreted. MT dependence could be tested using selective microtubule perturbations that preserve gross physiology. ¹³

Testing OR independently of the brain. Penrose's physics deserves a separate experimental track because biology is an unnecessarily complicated environment in which to discover a new law of quantum mechanics. The clean prediction is the scaling of collapse lifetime with gravitational self-energy. BEC spatial superpositions and increasingly massive mechanical superpositions can vary mass and separation while environmental decoherence is independently characterized. Howl, Penrose, and Fuentes explicitly developed such a BEC route. ⁶⁷

A positive gravity-collapse result would still not prove Orch OR. It would establish that Penrose supplied a real physical mechanism that biology *could* potentially exploit; microtubule coherence, orchestration, anesthesia sensitivity, and consciousness coupling would remain separate empirical questions. Conversely, a sufficiently broad set of experiments excluding the Penrose scaling over the parameter range required by realistic Orch OR calculations would remove the theory's defining "OR" mechanism even if interesting microtubule quantum biology survived.

Methodologically, all of these programs should adopt preregistration, shared raw data, prespecified signal-processing pipelines, blinding where possible, positive and negative reference materials, and replication by groups with no intellectual investment in Orch OR. The present literature is small enough that investigator dependence matters. A field making unusually large claims benefits disproportionately from unusually transparent experiments.

Best past experiments, evidence tables, and primary-source links

The following ranking is by **diagnostic scientific value**, not by whether the result favors Orch OR. Negative constraints are as valuable as positive observations when they sharply discriminate theories.

Rank by diagnostic value	Experiment	Brief method	Outcome	Reliability / interpretation
Donadi et al., 2021	Ultra-low-background germanium radiation measurement at Gran Sasso	Searched for spontaneous radiation generated by stochastic gravity-related collapse	Expected radiation absent; natural parameter-free DP version ruled out	High reliability, high relevance to a particular OR implementation. Does not exclude every possible Penrose OR completion. ⁴⁵
Kalra et al., 2023	Time-resolved Trp autofluorescence + quenching in tubulin/MTs	Inferred excitation diffusion and tested anesthetic perturbation	~6.6 nm MT diffusion; etomidate/ isoflurane reduce it	Strongest direct molecular Orch-linked experiment. Does not directly measure long-lived coherence or entanglement; specificity needs replication. ⁹
Li et al., 2025	Randomized/ blinded prolonged MT-drug treatment in male mice	Full incremental isoflurane LORR curves	EpoD/vinblastine increase sensitivity; paclitaxel slight resistance	Important independent constraint because it challenges a simple stabilizer-resistance prediction. Known control/physiology limitations. ⁷¹

Rank by diagnostic value	Experiment	Brief method	Outcome	Reliability / interpretation
Khan et al., 2024	Acute EpoB in rats, repeated 4% isoflurane LORR	Within-subject behavioral effect	+69 s, $d = 1.9$, N=8	Large interesting effect, small sample and indirect mechanism. Same-program replication exists in mice. ⁵²
Huang et al., 2026	Acute EpoB mouse LORR	Within-subject next-day test	+29 s, $d = 0.8$	Useful cross-species confirmation, but not independent laboratory replication. ³⁹
Babcock et al., 2024	Collective quantum-optical model + fluorescence quantum yield	Compared hierarchical tubulin structures	QY behavior consistent with collective superradiant architecture	Strong adjacent physics, but neither consciousness nor long-lived entanglement is measured. ²⁹
Emerson et al., 2013	Photoactive anthracene anesthetic + tubulin biochemistry + tadpole anesthesia	Localized binding and manipulated MT stability	β -tubulin implicated; MT stabilizers shift behavioral anesthetic response	Good causal molecular evidence, but agent/system is not equivalent to clinical volatile hypnosis. ³⁵
Li et al., 2018 xenon isotopes	80 randomized male mice; isotope-dependent xenon LORR	Compared spin-zero vs nuclear-spin xenon isotopes	Spin-bearing isotopes less potent	Intriguing quantum-sensitive pharmacology, but molecular substrate unknown and background isoflurane used. ⁴²
Kerskens & López Pérez, 2022	Human-brain ZQC MRI	Specialized pulse sequence intended as entanglement witness	Consciousness-associated signals reported	Potentially transformative but disputed. No MT localization or independent exact replication. ¹³

Rank by diagnostic value	Experiment	Brief method	Outcome	Reliability / interpretation
Saxena et al., 2020	Multi-scale impedance/resonance measurements	Tubulin, MT, neuron resonance spectra	Self-similar triplet structure reported	Interesting exploratory observation; low present diagnostic certainty because resonance is not a quantum witness and independent replication is absent.

50

For use in a claims database, the following compact evidence matrix separates the most easily conflated propositions.

Evidence item	Demonstrates ordinary MT function?	Demonstrates collective quantum/optical effect?	Demonstrates coherent superposition?	Demonstrates entanglement?	Links causally to anesthesia?	Demonstrates OR
Sahu electrical/optical MT work	Yes	Possibly/partly	No	No	No	N
Saxena resonance	Yes	Not necessarily	No	No	No	N
Kalra exciton migration	Yes	Yes, electronic collective transport	Not established	No	Direct molecular association	N
Babcock superradiance	Yes	Yes	Collective eigenstates theoretically; long-lived cognitive coherence not shown	No direct witness	No	N
Khan/Huang EpoB	Yes	Not measured	No	No	Yes, pharmacological causal link to LORR	N

Evidence item	Demonstrates ordinary MT function?	Demonstrates collective quantum/optical effect?	Demonstrates coherent superposition?	Demonstrates entanglement?	Links causally to anesthesia?	Demonstrates OR
Li EpoD/ paclitaxel/ vinblastine	Yes	Not measured	No	No	Yes, heterogeneous causal link	N
Kerskens ZQC MRI	N/A	Claimed nonclassical brain process	Possibly under interpretation	Claimed, disputed	No	N
Donadi Gran Sasso	N/A	N/A	N/A	N/A	N/A	Constrained one D collapse mode

The primary sources most worth linking directly from a claims site are:

Topic	Primary source and link
Original Orch OR formulation	Hameroff & Penrose, 1996, <i>Mathematics and Computers in Simulation</i> , DOI 10.1016/0378-4754(96)80476-9 . ⁷²
Penrose gravity-induced reduction	Penrose, 1996, <i>General Relativity and Gravitation</i> , DOI 10.1007/BF02105068 . ⁷³
Major revised Orch OR paper	Hameroff & Penrose, 2014, <i>Physics of Life Reviews</i> , DOI 10.1016/j.plev.2013.08.002 . ²³
Decoherence critique	Tegmark, 2000, <i>Physical Review E</i> , DOI 10.1103/PhysRevE.61.4194 . ⁷⁴
Decoherence response	Hagan, Hameroff & Tuszynski, 2002, <i>Physical Review E</i> , DOI 10.1103/PhysRevE.65.061901 . ⁴⁸
MT electronic-energy migration	Kalra et al., 2023, <i>ACS Central Science</i> , DOI 10.1021/acscentsci.2c01114 . ⁹
MT/Trp superradiance	Babcock et al., 2024, <i>Journal of Physical Chemistry B</i> , DOI 10.1021/acs.jpcc.3c07936 . ⁵¹
Single-MT water/electronic study	Sahu et al., 2013, <i>Biosensors and Bioelectronics</i> , DOI 10.1016/j.bios.2013.02.050 . ⁷⁵
Single-MT memory switching	Sahu et al., 2013, <i>Applied Physics Letters</i> , DOI 10.1063/1.4793995 . ⁷⁶
Fractal electromagnetic resonance	Saxena et al., 2020, <i>Fractal and Fractional</i> , DOI 10.3390/fractalfract4020011 . ⁵⁰
Tubulin/anesthetic anthracene experiment	Emerson et al., 2013, <i>JACS</i> , DOI 10.1021/ja311171u . ³⁵

Topic	Primary source and link
Rat epothilone-B anesthesia experiment	Khan et al., 2024, <i>eNeuro</i> , DOI 10.1523/ENEURO.0291-24.2024 . ⁵²
Independent MT-drug mouse experiment	Li et al., 2025, <i>BMC Anesthesiology</i> , DOI 10.1186/s12871-025-02956-9 . ⁵³
Mouse acute-EpoB follow-up	Huang et al., 2026, <i>Neuropharmacology</i> , DOI 10.1016/j.neuropharm.2026.110834 . ³⁹
Xenon isotope nuclear-spin experiment	Li et al., 2018, <i>Anesthesiology</i> , DOI 10.1097/ALN.0000000000002226 . ⁴²
Claimed nonclassical brain MRI signal	Kerskens & López Pérez, 2022, <i>Journal of Physics Communications</i> , DOI 10.1088/2399-6528/ac94be . ⁴³
Gran Sasso DP-collapse constraint	Donadi et al., 2021, <i>Nature Physics</i> , DOI 10.1038/s41567-020-1008-4 . ⁴⁵
Orch OR implications of collapse constraint	Derakhshani et al., 2022, <i>Physics of Life Reviews</i> , DOI 10.1016/j.plrev.2022.05.004 . ⁷⁷
Qualification of that claimed refutation	McQueen, 2023, <i>Physics of Life Reviews</i> , DOI 10.1016/j.plrev.2023.01.021 . ⁷⁸
Proposed BEC test of gravitational collapse	Howl, Penrose & Fuentes, 2019, <i>New Journal of Physics</i> , DOI 10.1088/1367-2630/ab104a . ⁶⁷
Strong GABA _A anesthetic genetic control	Jurd et al., 2003, <i>FASEB Journal</i> , DOI 10.1096/fj.02-0611fje . ⁵⁷

The strongest overall research strategy is thus a **two-front falsification program**. On the biology side, demonstrate a precisely defined nonclassical microtubule state, causally connect it to conscious-state transitions, and use target-level genetics to prove that anesthetics act on that state. On the physics side, test Penrose's E_G -dependent collapse independently in clean mesoscopic superpositions. Orch OR becomes compelling only if those programs ultimately meet: a real gravitational collapse law plus a verified biological system that exploits it. At present, neither half has crossed that evidential threshold, but the microtubule/anesthesia half has progressed far enough that carefully designed independent experiments are scientifically justified rather than merely speculative. ⁷⁹

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